

ASSIGNMENT-5

1. Given a range of first 10 numbers, Iterate from start number to the end number and print the sum of the current number and previous number.

```
In [4]: start = 1
end = 10

previous = start

for i in range(start, end + 1):
    if i == start:
        print(i)
    else:
        print(i + previous)
        previous = i
```

```
1
3
5
7
9
11
13
15
17
19
```

2. Given a string, display only those characters which are present at an even index number.

```
In [3]: s = input("Enter a string: ")

for i in range(len(s)):
    if i % 2 == 0:
        print(s[i])
```

```
S
m
n
N
y
k
```

3. Given a string and an integer number n, remove characters from a string starting from zero up to n and return a new string.

```
In [6]: s = input("Enter a string: ")
n = int(input("Enter n: "))

new_string = s[n:]

print("New string:", new_string)
```

New string: icon

4. Given a list of numbers, return True if first and last number of a list is same.

```
In [7]: numbers = [10, 20, 30, 40, 10]

if numbers[0] == numbers[-1]:
    print("True")
else:
    print("False")
```

True

5. Given a list of numbers, Iterate it and print only those numbers which are divisible of 5.

```
In [8]: numbers = [10, 15, 22, 30, 41, 50, 63]
for i in numbers:
    if i % 5 == 0:
        print(i)
```

10
15
30
50

6. Return the total count of sub-string appears in the given string.

```
In [13]: string = input("Enter a string: ")
substring = input("Enter substring: ")

count = string.count(substring)

print("Total count:", count)
```

Total count: 1

7. Given a two list of numbers create a new list such that new list should contain only odd numbers from the first list and even numbers from the second list.

```
In [9]: list1 = [1, 2, 3, 4, 5, 6]
list2 = [10, 11, 12, 13, 14, 15]

new_list = []

for i in list1:
    if i % 2 != 0:
        new_list.append(i)

for i in list2:
    if i % 2 == 0:
        new_list.append(i)

print("New list:", new_list)
```

New list: [1, 3, 5, 10, 12, 14]

8. Given a list iterate it and display numbers which are divisible by 5 and if you find number greater than 150 stop the loop iteration.

```
In [17]: numbers = [10, 25, 40, 55, 100, 125, 160, 200]

for i in numbers:
    if i > 150:
        break

    if i % 5 == 0:
        print(i)
```

10
25
40
55
100
125

9. Use a loop to display elements from a given list which are present at even positions.

```
In [19]: numbers = [10, 20, 30, 40, 50, 60, 70, 80]

for i in range(len(numbers)):
    if i % 2 == 0:
        print(numbers[i])
```

10
30
50
70

10. Find the sum of the series 2 + 22 + 222 + 2222 + ... n terms.

```
In [21]: n = int(input("Enter number of terms: "))

number = 0
sum = 0

for i in range(n):
    number = number * 10 + 2
    sum = sum + number

print("Sum =", sum)
```

Sum = 246

11. Given a two list. Create a third list by picking an odd-index element from the first list and even index elements from second.

```
In [23]: list1 = [10, 20, 30, 40, 50]
list2 = [60, 70, 80, 90, 100]

list3 = []
```

```

for i in range(len(list1)):
    if i % 2 != 0:
        list3.append(list1[i])

for i in range(len(list2)):
    if i % 2 == 0:
        list3.append(list2[i])

print("Third list:", list3)

```

Third list: [20, 40, 60, 80, 100]

12. Given an input list removes the element at index 4 and add it to the 2nd position and also, at the end of the list.

```

In [25]: numbers = [10, 20, 30, 40, 50, 60]

element = numbers.pop(4)

numbers.insert(2, element)
numbers.append(element)

print(numbers)

```

[10, 20, 50, 30, 40, 60, 50]

13. Given a list slice it into a 3 equal chunks and reverse each list.

```

In [27]: numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9]
n = len(numbers) // 3

list1 = numbers[:n]
list2 = numbers[n:2*n]
list3 = numbers[2*n:]

list1.reverse()
list2.reverse()
list3.reverse()

print("Chunk 1:", list1)
print("Chunk 2:", list2)
print("Chunk 3:", list3)

```

Chunk 1: [3, 2, 1]

Chunk 2: [6, 5, 4]

Chunk 3: [9, 8, 7]

14. Given a two list of equal size create a set such that it shows the element from both lists in the pair.

```

In [29]: list1 = [1, 2, 3, 4, 5]
list2 = [3, 4, 5, 6, 7]

set1 = set(list1)
set2 = set(list2)

```

```
common = set1 & set2
print("Common elements:", common)
```

Common elements: {3, 4, 5}

15. Given a string of odd length greater 7, return a string made of the middle three chars of a given String.

```
In [31]: s = input("Enter a string: ")
middle = len(s) // 2
print("Middle three characters:", s[middle-1:middle+2])
```

Middle three characters: uma

16. Given 2 strings, s1 and s2, create a new string by appending s2 in the middle of s1.

```
In [33]: s1 = input("Enter first string: ")
s2 = input("Enter second string: ")

middle = len(s1) // 2
new_string = s1[:middle] + s2 + s1[middle:]

print("New string:", new_string)
```

New string: SumMohantyan

17. Given 2 strings, s1, and s2 return a new string made of the first, middle and last character of each input string.

```
In [35]: s1 = input("Enter first string: ")
s2 = input("Enter second string: ")

m1 = len(s1) // 2
m2 = len(s2) // 2

new_string = s1[0] + s1[m1] + s1[-1] + s2[0] + s2[m2] + s2[-1]
print("New string:", new_string)
```

New string: SymSvm

18. Arrange string characters such that lowercase letters should come first.

```
In [37]: s = input("Enter a string: ")

lower = ""
other = ""

for i in s:
    if i.islower():
        lower = lower + i
    else:
        other = other + i

print("New string:", lower + other)
```

New string: ivamahooS S

19. Count all lower case, upper case, digits, and special symbols from a given string.

```
In [39]: s = input("Enter a string: ")

lower = 0
upper = 0
digit = 0
special = 0

for i in s:
    if i.islower():
        lower = lower + 1
    elif i.isupper():
        upper = upper + 1
    elif i.isdigit():
        digit = digit + 1
    else:
        special = special + 1

print("Lowercase:", lower)
print("Uppercase:", upper)
print("Digits:", digit)
print("Special symbols:", special)
```

```
Lowercase: 13
Uppercase: 2
Digits: 0
Special symbols: 1
```

20. Given a string, return the sum and average of the digits that appear in the string, ignoring all other characters.

```
In [11]: s = input("Enter a string: ")

sum = 0
count = 0

for i in s:
    if i.isdigit():
        sum = sum + int(i)
        count = count + 1

if count > 0:
    average = sum / count
else:
    average = 0

print("Sum =", sum)
print("Average =", average)
```

```
Sum = 6
Average = 2.0
```

21. Write a Python program to remove duplicates from a list.

```
In [43]: numbers = [10, 20, 10, 30, 20, 40, 30]

new_list = []

for i in numbers:
    if i not in new_list:
        new_list.append(i)

print("List after removing duplicates:", new_list)
```

List after removing duplicates: [10, 20, 30, 40]

22. Write a Python program to find the second largest number in a list.

```
In [45]: numbers = [10, 50, 30, 80, 20, 60]
numbers.sort()
print("Second largest number:", numbers[-2])
```

Second largest number: 60

23. Write a Python program to sort a list of elements using the bubble sort algorithm.

```
In [47]: numbers = [5, 2, 8, 1, 3]

n = len(numbers)

for i in range(n):
    for j in range(0, n - i - 1):
        if numbers[j] > numbers[j + 1]:
            temp = numbers[j]
            numbers[j] = numbers[j + 1]
            numbers[j + 1] = temp

print("Sorted list:", numbers)
```

Sorted list: [1, 2, 3, 5, 8]

24. Write a Python program to find the common elements between two lists.

```
In [49]: list1 = [10, 20, 30, 40, 50]
list2 = [30, 40, 50, 60, 70]

common = []

for i in list1:
    if i in list2:
        common.append(i)

print("Common elements:", common)
```

Common elements: [30, 40, 50]

25. Write a Python program to count the frequency of each element in a list.

```
In [51]: numbers = [10, 20, 10, 30, 20, 10, 40]
```

```

checked = []

for i in numbers:
    if i not in checked:
        count = 0

        for j in numbers:
            if i == j:
                count = count + 1

        print(i, "=", count)
        checked.append(i)

```

```

10 = 3
20 = 2
30 = 1
40 = 1

```

26. Write a Python program to count the number of strings where the string length is 2 or more and the first and last character are same from a given list of strings.

```

In [12]: words = ['abc', 'xyz', 'aba', '1221', 'z']
count = 0

for word in words:
    if len(word) >= 2 and word[0] == word[-1]:
        count += 1

print("Count:", count)

```

Count: 2

27. Write a Program that get two lists as input and check if they have at least one common member

```

In [13]: list1 = [1, 2, 3, 4, 5]
list2 = [5, 6, 7, 8, 9]

has_common = False

for item in list1:
    if item in list2:
        has_common = True
        break

print("Have at least one common member:", has_common)

```

Have at least one common member: True

28. Write a Python program to print a specified list after removing the 0th, 4th and 5th elements.

```

In [57]: colors = ['Red', 'Green', 'White', 'Black', 'Pink', 'Yellow', 'Blue']

# Retain items whose index is NOT 0, 4, or 5
result = [colors[i] for i in range(len(colors)) if i not in (0, 4, 5)]

```



```
print("Modified List:", result)
```

Modified List: ['Green', 'White', 'Black', 'Blue']

29. Write a Python program to generate and print a list of first and last 5 elements where the values are square of numbers between 1 and 30.

```
In [59]: squares = [i**2 for i in range(1, 31)]

first_5 = squares[:5]
last_5 = squares[-5:]

print("First 5 squares:", first_5)
print("Last 5 squares:", last_5)
```

First 5 squares: [1, 4, 9, 16, 25]

Last 5 squares: [676, 729, 784, 841, 900]

30. Write a Python Program to print all Possible Combinations from the three Digits.

```
In [61]: d1, d2, d3 = 1, 2, 3
digits = [d1, d2, d3]

print("Possible combinations:")
for i in range(3):
    for j in range(3):
        for k in range(3):
            # Ensure each digit is used only once per combination
            if i != j and j != k and i != k:
                print(digits[i], digits[j], digits[k])
```

Possible combinations:

```
1 2 3
1 3 2
2 1 3
2 3 1
3 1 2
3 2 1
```